LS-CAT Management Board Meeting

Agenda

All times CDT

Location:	Bluejeans
11:00	Approval of minutes and agenda (JS)
11:10	SBC (KB)
	Office Space (KB)
11:20	Survey results, Future Directions (JS)
11:30	S10 Eiger2 (KB)
	S10 MD3-UP (KB)
11:40	LS-CAT FY 2020 Budget (KB)
11:50	Fall Meeting Agenda Planning
	Adjourn

Audio:
+1.408.740.7256
+1.408.317.9253
+1.408.317.9254

+1.312.216.0325
+1.866.226.4650 (Toll Free)

Meeting ID: 386 421 2738

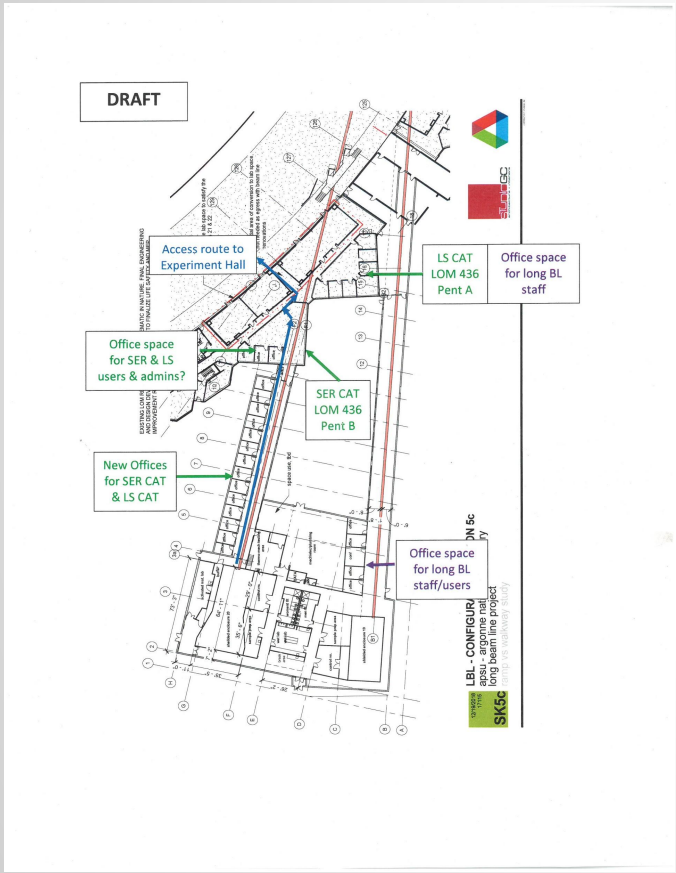
Video:
https://bluejeans.com/kbr781
Meeting ID: 386 421 2738

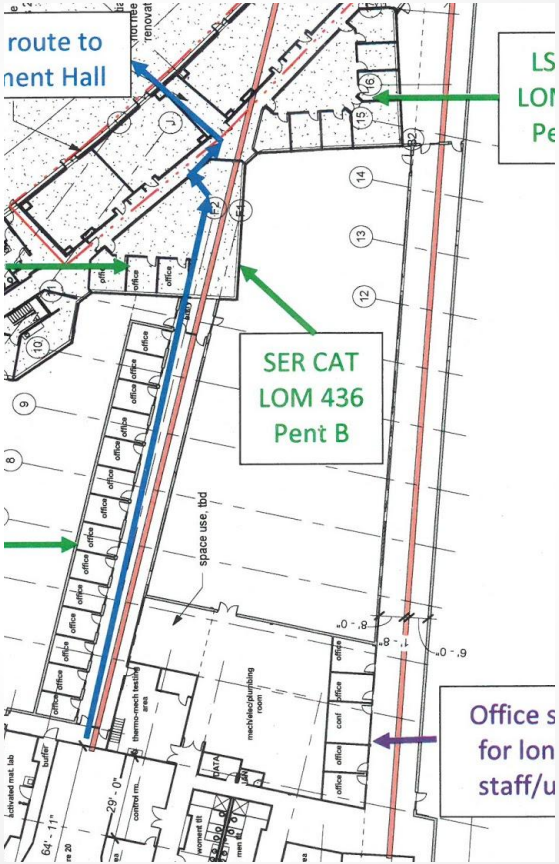
SBC as LS-CAT Member?

SBC was recently absorbed into the APS management structure. It had been part of ANL Biology Department. The still do not have an identified location on the experimental floor after the APS upgrade, however, they do not appear to in a hurry to resolve this issue.

There are advantages and disadvantages to having SBC as a member. Our best way forward is to minimize the time spent fretting about it and instead work on the LS-CAT Agreement so that there is an obvious path to include this sort of partnership. Recall that such an arrangement with SBC will likely NOT included cash but could include personnel and equipment.

LS-CAT Offices After Long Beamline Construction





Current plans call for offices to be constructed along the hallway that extends from the current SER-CAT offices to the planned long beamlines. SER-CAT and LS-CAT would share this space as well as the common area. All offices will be on the correct side of the beampipe.

Future Directions Committee Report

		Priority	Funded	Worst Case	Item	Est Cost	Contingency	Worst Case Contingency	Worst Case Spending
Total of all Costs	$9,548,000	1	1	1	Upgrade main storage to 1PB	40,000	10,000	10,000	50,000
Maximum Priority	11	2	1	1	Move Inboard Beam masks downstream and lengthen bellows	10,000	2,500	12,500	62,500
Priority Costs	$1,078,000	3	1	1	Remove side-bounce mono's water to vacuum braze (2 each)	30,000	7,500	20,000	100,000
		4	1	1	Remove commissioning window: install differential pump	10,000	2,500	22,500	112,500
Worst Case Cost Contingency	25%	5	1	1	Service Cryocooler (1 each)	5,000	1,250	23,750	118,750
		6	1	1	Service Cryo-Tigers (10 each)	75,000	18,750	42,500	212,500
Funding Level	$500,000	7	1	1	Service Cryo-Jets	5,000	1,250	43,750	218,750
Max priority w/contingency		8	1	1	Upgrade Motor Controllers	50,000	12,500	56,250	281,250
		9	1	1	Service D-Mono Electro-Mechanics	3,000	750	57,000	285,000
		10	1	0	Replace D diffractometer with MD3-UP	650,000	20,000	77,000	955,000
		11	1	0	Upgrade side-bounce monos	200,000	50,000	127,000	1,205,000
		12	0	0	New D-Mono Cooling	20,000	5,000	132,000	1,230,000
		13	0	0	New D-Mono Crystals	20,000	5,000	137,000	1,255,000
		14	0	0	Replace F Diffractometer with MD2S	460,000	20,000	157,000	1,735,000

S10 Proposal: Eiger2 X 16M



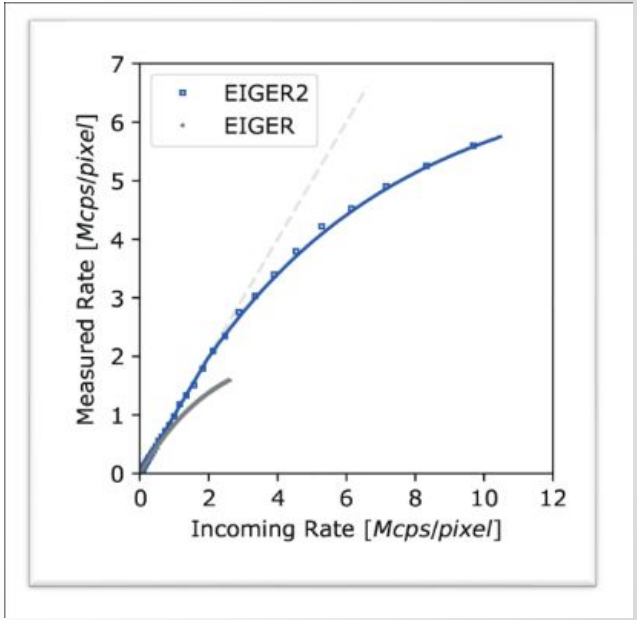
For LS-CAT there are three features of this detector that are particularly attractive:

1. Zero-deadtime photon counting increases by at least a factor 4 the maximum usable photon flux that impinges on the detector.

2. More linear count rate corrections meaning higher quality data.

3. Larger size that when coupled with the planned MD3-UP diffractometer means we'll be able increase the available solid angle at a given distance.

Together these would mean better statistics and higher resolution structures. It would be more likely that that one good crystal would yield results.



S10 Proposal: MD3-UP

Approximately an order of magnitude higher precision over our current MD-2 diffractometers.

Excellent match to the beams available after the APS upgrade.

Built-in support for popular data collection modes such as raster scanning, helical scans, and 96-well plate scans.

[Marketing Video]



FY 2019 Budget and Predicted FY 2020 Budget

FY19 Est Spending: Best guess of how things are going based on the first 8 months of the fiscal year (which started September 1)

FY19 Budget: Current budget as approved last spring

FY20 Proposed: The proposed budget for the upcoming fiscal year.

Description	FY19 Est Spending	FY19 Budget	FY20 Proposed
Academic Personnel	520,060	532,676	537,489
Non-Academic Personnel	105,441	101,906	110,547
Fringe Benefits	217,565	234,370	245,679
Salaries Restricted	228,914	220,784	223,165
Supplies	35,855	40,000	40,000
Supplies, Restricted	2,185	2,500	2,500
Services	15,485	30,000	30,000
Services, Restricted	17,733	20,000	20,000
External Space Rental	131,000	131,303	135,242
Travel, Domestic	787	5,000	5,000
Utilities & Telephone	7,558	9,000	9,000
Capital Equipment	127,423	21,959	34,140
Indirect Cost - Sponsored	289,982	389,413	398,316
Totals:	**1,699,989**	**1,738,911**	**1,791,078**

LS-CAT Budget for FY 2020

Salaries				100% KB, MB, JVO, SA, JSB, EK, ZW
	Base	871,201		50% MG
	Fringe	245,679		
	Total		1,116,879	
Items F&A Exempt				
	Capital Equip.	34,140		
	Rent	135,242		
	Telcom	9,000		
	Total		178,382	
Non-Salary Items Subject to F&A				
	Lab Supplies	40,000		
	Office Supplies	2,500		
	Lab Services	30,000		
	Insurance	20,000		
	Travel	5,000		
	Total		97,500	
F&A			398,316	
Total			1,791,078	
Per Share			358,216	NU UM MSU
Per 1/2 Share			179,108	UW UI VU
Per 1/4 Share			89,554	WSU VARI

Vote 1:
Approval of amounts shown.

Vote 2:
Exact amount of capital equipment needed depends on what breaks and which upgrades cannot be postponed. Monies left on this line will be added to capital fund or, conversely, when outlays exceed this amount we'll dip into the capital fund. As always, the board will be kept informed about unusually large expenses. Vote 2 allows the capital equipment funds to be used for normal upkeep of the facility.